

Set 17: Electron dot diagrams

Notes

When making observations about materials we are developing an awareness of the different properties materials have. The sorts of properties observed include colour, state, hardness, elasticity, conductivity, melting point and boiling point. Many of these properties can be explained by knowledge of the type of bonding that occurs in the material. Therefore, in order to understand the many materials that surround us it is essential to have an understanding of the structure of ionic, metallic, covalent network and covalent molecular substances. This will then enable us to develop an understanding of the relationships between the properties and structures of these substances.

The following sets of questions and problems are designed to get you thinking about the relationship between structure and properties of materials.

Electron dot diagrams

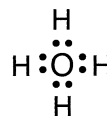
Electron dot diagrams are a really useful tool to help you understand the bonding arrangements of atoms. From these diagrams it is possible to determine the shape of molecules and also the polarity. They are quite easy to draw for simple molecules as long as you follow some simple guidelines. Try to satisfy a full outer level of eight electrons (the octet rule) by:

- Determining the total number of valence electrons from all the atoms involved;
- Using a pair of electrons to form a single bond between each pair of bonded atoms; If there are not enough electrons, consider a multiple covalent bond, if there are too many electrons, consider a coordinate covalent bond;
- Placing the remaining electrons to satisfy the octet rule. Be sure the electrons are drawn in pairs.

Example 1: Methane CH_4

Carbon has four electrons and each of the hydrogens has one electron, giving a total of 8 valence electrons.

- Trying a single covalent bond for each atom

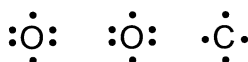


A single bond between each carbon and hydrogen will satisfy all the elements.

Example 2: Carbon dioxide CO_2

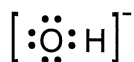
Each oxygen has six electrons and carbon has four, giving a total of 16 valence electrons.

- Try a single covalent bond for each atom
Single bonds will not satisfy the carbon, there are not enough electrons.
- If there are not enough electrons, consider a multiple covalent bond. A double bond between each oxygen and carbon will satisfy all the elements.
- Place the remaining electrons to satisfy the octet rule. Be sure the electrons are drawn in pairs.



Example 3: Hydroxide ion OH⁻

Oxygen has six electrons and hydrogen has one. As a negative one ion, an extra electron has been gained, giving a total of 8 valence electrons.



A single bond between each oxygen and hydrogen will satisfy the elements, with the extra electron completing an octet for oxygen and is shown as a -1 charge on the ion.

Example 4: Ionic substances

Square brackets are used when representing ions in electron dot diagrams of ions and ionic substances.

E.g. in NaCl, sodium has one valence electron and when forming an ionic bond becomes a positive ion by losing its valence electron.

Chlorine has seven valence electrons and accepts an electron when forming ionic bonds becoming a negative ion.



sodium atom

chloride atom

sodium chloride

Set 17: Exercises

- Draw electron dot diagrams for the following atoms and ions:
 - Na
 - Br
 - P
 - S²⁻
 - H⁺
 - N³⁻
- Draw electron dot diagrams for the following covalent molecular compounds (The central atom is in **bold** type) :
 - H₂**O**
 - C**H₃Cl
 - P**H₃
 - H₂**S**
 - C**HCl₃
 - HCN
- Draw electron dot diagrams for the following ionic compounds:
 - NaOH
 - CaCl₂
 - Fe₂O₃
 - K₂S
 - Mg(OH)₂
 - AgNO₃
- Explain how drawing an electron dot diagram can help you to predict the solubility of a substance.

Notes